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隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

This study measures the difference between the transaction value digital platforms process in Indonesia and the revenue those platforms recognize for themselves. I call that difference the invisible wedge. The absolute wedge is $W = V - R$, where V is transaction value and R is platform revenue; the Ecosystem Ratio, $E = (V - R)/R$, expresses the same boundary relative to revenue and is constructed for this study rather than adopted from prior work. Because no Indonesian platform discloses a strictly country-labelled matched pair across the study period, observations are retained under named evidence tiers rather than pooled as if equal in scope. The inventory spans seventeen candidate platform-year levels from FY2019 to FY2025. The proposal asks how large the wedge is, how it changes, and what explains those changes, then uses official statistics and Indonesia's marketplace-reporting rules to locate related activity and records beyond platform revenue.

1. Introduction

Southeast Asia's digital economy grew from US\$100 billion in gross transaction value in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024; reported there as gross merchandise value). Much of that activity is coordinated through digital platforms, but transaction value can be far larger than the revenue a platform recognizes for itself. Platforms record transactions, determine participant payouts, and book their own revenue. I call the difference between transaction value processed through a platform and platform-booked revenue the invisible wedge.

Suppose a platform processes 100 units of transaction value but recognizes only 10 units as revenue. The remaining 90 units are still recorded within the platform's system but do not constitute platform revenue; they may include merchant receipts, driver payouts, inventory costs, taxes paid elsewhere, and other pass-through payments. The paper measures this accounting gap without assuming that the residual is profit, taxable income, or unpaid tax. The 90-unit difference is the absolute wedge, $W = V - R$.

I measure the wedge relative to revenue with the Ecosystem Ratio, $E = (V - R)/R$. For FY2023, three selected platform cases imply a combined wedge of approximately US\$40.07 billion, about 2.9 percent of Indonesia's 2023 nominal GDP. This is a scale comparison only: the wedge is transaction value outside platform revenue, not GDP value added, participant profit, or a tax gap.

The term invisible therefore does not imply concealment. A platform may hold seller- or worker-linked transaction records even when those records do not enter routine third-party income reporting. The wedge does not measure that reporting gap. It measures the transaction-revenue boundary; the reporting question is whether records behind that activity are separately transmitted and usable outside the platform.

1.1 Research question and objectives

One question drives the proposal:

How large is the invisible wedge in Indonesia, how does it change over time, and what explains those changes?

The empirical design follows three objectives: measure the wedge longitudinally under explicit evidence tiers; explain material movements using issuer-disclosed revenue components; and connect the measured boundary to wider Indonesian activity and to seller-linked reporting institutions. A separate 48-issuer-year module outside Indonesia is retained only as external corroboration, not as a second main sample or research question.

1.2 Contributions

First, longitudinal measurement. The evidence base expands from a one-year comparison to candidate observations spanning FY2019-FY2025, while keeping direct Indonesia-aligned, direct scope-pending, and conditional observations separate.

Second, mechanism. The thesis reconciles material transaction-revenue divergences against disclosed monetization, customer incentives, gross-versus-net recognition, and business perimeter. These are arithmetic reconciliations, not causal decompositions.

Third, economic and institutional interpretation. Evidence from BPS-Statistics Indonesia (Badan Pusat Statistik) locates wider e-commerce activity and recordkeeping beyond issuer accounts, while Indonesia's Regulation PMK 37/2025 (Peraturan Menteri Keuangan) and Directorate General of Taxes (DJP) materials show how platform-held seller and transaction records can enter reporting and withholding. The proposal does not claim a completed compliance or tax-revenue effect.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy problem. Platform accounts make the intermediation boundary observable, but that boundary can change the conclusions investors and researchers draw about scale, growth and monetization. De Franco, Kothari and Verdi (2011) motivate the need to keep accounting bases comparable, while Berg et al. (2020) show that intermediary-held digital traces can contain economically useful information even when conventional records are thin. The proposal combines these ideas by treating transaction value and recognized revenue as related but non-equivalent measures of the same platform ecosystem.

2. Literature Review and Research Gap

2.1 Platform economics and revenue recognition

Multisided-platform theory explains why a platform's revenue need not move proportionally with the transaction value it coordinates (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016). Accounting further determines how much facilitated commerce becomes recognized revenue: principal-agent treatment, customer incentives, service mix, acquisitions, and reporting perimeter can move revenue without an equivalent change in underlying commerce. De Franco, Kothari and Verdi (2011) show why financial-statement comparability matters when users interpret such differences.

2.2 Informality, digital records, and third-party information

Platform participants occupy an unusual position. Their transactions can be priced, recorded, and settled through a formal intermediary while participants remain self-employed, unregistered, below filing thresholds, or outside automatic reporting. The object measured here is therefore not the shadow economy as conventionally defined (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Digital records can nevertheless carry economic information even when conventional records are thin (Berg et al. 2020), and platform-mediated work leaves observable financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

A central public-finance result is that third-party information can materially change compliance and enforcement (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules and the European Union's DAC7 regime provide direct precedents for using platform-held seller information in reporting. Indonesia's PMK 37/2025 applies the same broad logic through seller identification, transaction-linked reporting, and marketplace withholding.

2.3 Measuring the digital economy

Digital-economy measurement frameworks address the same boundary from the national-accounting side. Official guidance separates the underlying buyer-seller transaction from the intermediation service the platform provides, and rejects treating gross transaction value as value added or as an alternative output measure (Ahmad and Schreyer 2016; International Monetary

Fund 2018; OECD 2023; United Nations et al. 2025). Digitalisation also creates practical difficulties of classification, business perimeter, cross-border activity and revision of private-source data. This literature establishes what the wedge is not, and is the reason the proposal treats V - R as an accounting boundary rather than as unmeasured output.

2.4 Research gap

The literature explains why transaction value can exceed platform revenue, why accounting can move the two apart, and why third-party records matter. What remains missing is a longitudinal, source-auditable account for one market that measures the wedge, explains why it moves, and then asks where the wider activity and its records appear beyond platform revenue.

3. Theoretical Framework and Key Variables

Let V denote transaction value and R platform-recognized revenue for a matched platform scope and period. The core measures are:

$$W = V - R \quad E = (V - R) / R \quad D = g(V) - g(R)$$

W is the absolute invisible wedge; E is the Ecosystem Ratio, which is my own construction for this paper rather than a measure adopted from a prior study; and D is the annual growth divergence between transaction value and revenue. A higher E indicates more transaction value outside platform revenue per unit of platform revenue; it does not by itself imply more participant profit, taxable income, value added, non-compliance, or tax due.

Gross transaction value (GTV) is used as the proposal's common label for the value processed through a platform. Issuer terminology differs: Sea Limited and Blibli use GMV, Grab uses GMV for deliveries and TPV for financial services, and GoTo uses GTV. These source labels and business perimeters remain recorded in the empirical files rather than being treated as economically interchangeable. Accordingly, the main longitudinal comparisons are within named series; cross-platform levels are descriptive and scope-labelled.

3.1 Hypotheses

The framework yields five hypotheses, which Table 1 states together with their current status. H1 to H3 and H5 are addressed by the preliminary evidence in Section 5 and are reported as current status rather than settled results; H4 is the principal question the thesis proposes to take further. None is framed as a causal claim.

Table 1. Hypotheses and current evidence status.

Hypothesis	Status in the current evidence
H1 Transaction value and platform revenue can diverge within a series.	Supported in the candidate inventory: 12 transitions, 2 opposite-sign, median absolute divergence 42.02 pp. Final Indonesian admission unresolved.
H2 Aggregate e-commerce growth partly reflects more businesses rather than more value per business.	Supported for 2023-2024: value +17.08%, businesses +15.31%, implied value per business +1.54%. Arithmetic decomposition, not causal entry.
H3 Marketplace participation is associated with financial recordkeeping.	Associated in BPS published business-level evidence; the within-province change is unstable ($r = 0.309$, $p = 0.066$). Not causal.
H4 Platform-held records are not yet administratively linked to seller obligations.	Not yet tested. Requires implementation and identifier evidence, not legal architecture alone. This is the question the thesis proposes to advance.
H5 Measurement choices can alter conclusions.	Supported by the reporting-vintage, revenue-definition and business-perimeter modules. Consequence assessed comparison by comparison.

4. Data and Methodology

4.1 Admission rules and evidence tiers

A platform-period is eligible only when transaction value and revenue cover the same period; geography and business scope can be evaluated; units and definitions are known; derived inputs trace to source; and structural breaks or revised reporting bases are flagged. Repeated publication vintages are not counted as independent observations. Eligibility does not imply final main-sample admission.

Table 2. Evidence tiers for Indonesian platform observations. Tiers differ in how directly each series maps to Indonesia; they are never pooled into a single monetary total.

Evidence tier	Series	Levels	Transitions
Direct Indonesia-aligned	Tokopedia e-commerce	2	1
Direct, scope-pending	Blibli 3P; Bukalapak Group	9	7
Conditional reconstruction	Grab; Shopee Indonesia	6	4
All tiers (diagnostic)	Five series	17	12

Table 2 shows that only two of the seventeen retained levels are directly Indonesia-aligned, so sample size and evidence quality move in opposite directions. Tokopedia is the strongest Indonesia-aligned segment but is not a literal country line. Blibli includes online travel, Bukalapak reports at Group scope with overseas operations, and Grab/Shopee require a derived or external country component. Tokopedia FY2021 and Bukalapak FY2024 are excluded because transaction and revenue periods do not match.

Three counts are intentionally different. The all-tier inventory contains 17 retained levels; the executed direct tiers contain 11 levels (2 Indonesia-aligned and 9 scope-pending); and the broader direct-candidate sensitivity contains 13 periods, including a Blibli FY2020 prospectus observation, yielding 9 annual transitions. These counts answer different construction questions and are therefore not collapsed into one N.

4.2 Comparability strategy

The main longitudinal comparisons are therefore made within a named series rather than by treating all platform levels as economically interchangeable. A within-series transition holds the issuer, business perimeter and disclosure convention as constant as the available filings permit; changes in transaction value and revenue can then be interpreted as movements in that series rather than as differences created purely by cross-company accounting definitions.

Cross-platform levels still matter, but for a different purpose. The FY2023 cross-section is descriptive evidence of scale under explicitly labelled evidence classes, not a pooled estimator of a national total. Likewise, the external 48 issuer-years are corroboration that the transaction-revenue boundary is not unique to Indonesia. This hierarchy is why the proposal can use heterogeneous disclosures without pretending that they form one homogeneous panel.

4.3 Conditional country evidence

Grab discloses Indonesia revenue but not Indonesia transaction value. Its conditional country transaction value is therefore derived as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The construction assumes the Group monetization rate applies to Indonesia and is treated as sensitivity evidence, not direct disclosure. Shopee's Indonesia transaction value uses Momentum Works' recurring Southeast Asian e-commerce estimates because Sea Limited does not disclose an Indonesian country figure; it is labelled third-party wherever used.

5. Preliminary Evidence and Feasibility

5.1 FY2023 scale

Table 3. Indonesia-focused FY2023 cross-section, US\$ billions. V is transaction value, R platform revenue, W the wedge and E the Ecosystem Ratio. The final row sums three documented cases under mixed evidence classes and is not a national total.

Case	V	R	W	E	Evidence class
Grab Indonesia	5.381	0.605	4.776	7.895x	Derived V
Tokopedia	16.331	0.405	15.926	39.296x	Direct pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Derived V and R
Selected platforms	43.233	3.162	40.070	12.671x	Sum, not national total

Table 3 reports the FY2023 cross-section. These cases establish the order of magnitude and feasibility of the measure. Sensitivity and leave-one-platform-out checks do not remove the large transaction-revenue gap, but they are treated as robustness material rather than the thesis's main result.

5.2 Longitudinal movement and mechanism

Table 4. Within-series annual divergence under two admission rules. Counts differ because evidence admission differs; both are reported rather than resolved into one headline.

	All-tier inventory	Direct-candidate sensitivity
Transitions	12	9
Revenue grows faster	10	6
Transaction grows faster	2	3
Opposite-sign transitions	2	3
Median absolute divergence	42.02 pp	42.94 pp

Table 4 compares the two admission rules. The summaries differ because the rules differ; neither is a filtered restatement of the other. Tokopedia FY2022-FY2023 provides the cleanest mechanism example: transaction value falls 8.90 percent while selected third-party net segment revenue rises 53.20 percent. Of the arithmetic increase in net revenue, 60.56 percent is associated with lower customer incentives and 39.44 percent with higher gross revenue. This reconciles the divergence without claiming causality.

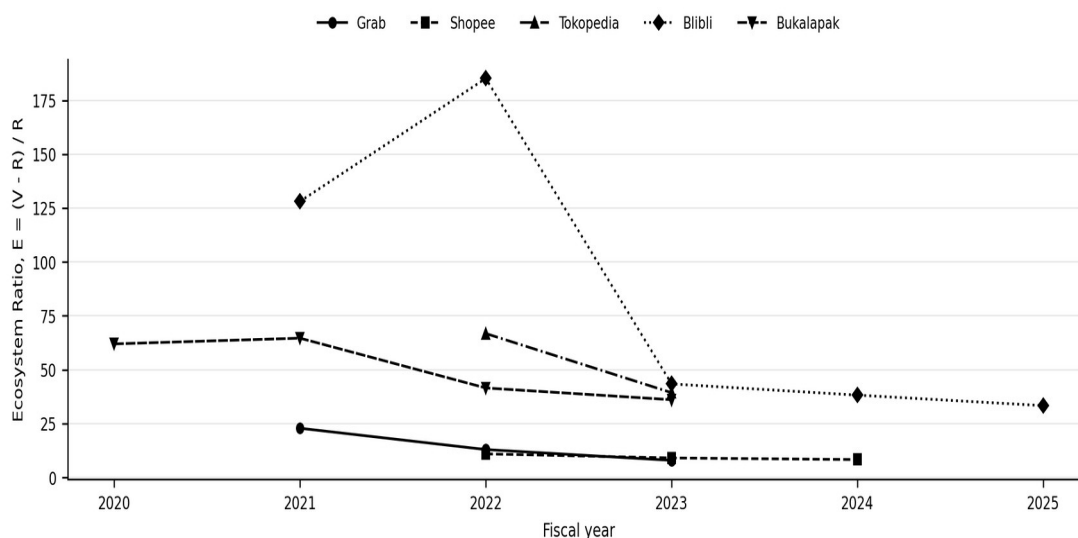


Figure 1. Ecosystem Ratio across the extended candidate series. Each line follows one disclosed or transparently reconstructed series through time; levels are shown by series and are not pooled across evidence tiers.

Figure 1 makes two features of the extended sample visible that the summary table alone does not. First, the wedge persists across every retained series, although its magnitude differs substantially by business model and reporting perimeter. Second, several series show large changes in the ratio over time, which is why the thesis focuses on explaining movements rather than treating one cross-sectional ratio as a fixed platform characteristic. The figure is descriptive: differences between lines are not interpreted as causal platform effects.

5.3 Wider Indonesian activity and reporting

BPS reports national e-commerce transaction value rising 17.08 percent from 2023 to 2024. Over the same period, the directly reported marketplace component rises only 1.45 percent, while the non-marketplace component rises 20.57 percent; approximately 98.46 percent of the nominal increase therefore falls outside the marketplace component. These aggregates provide a wider view of e-commerce activity and are not reconciled one-for-one to issuer transaction value.

PMK 37/2025 provides the institutional bridge. The verified implementation sequence records marketplace designation on 1 July 2026, collection effective 1 August, postponement through 31 October, and scheduled implementation on 1 November 2026. The regulation is used as evidence of seller-linked reporting architecture, not as proof of prior underpayment or a completed compliance effect.

Outside Indonesia, 48 matched issuer-years across eight platform businesses show that transaction value and revenue can also diverge under other business models. This evidence remains external corroboration only; it is neither pooled with Indonesia nor treated as a representative global sample.

5.4 Payment traces and administrative records

Bank Indonesia's payment-system statistics give an independent view of digital trace activity. Between 2023 and 2024, BPS e-commerce transaction value grows 17.08 percent, while electronic-money shopping value grows 30.47 percent, mobile-banking payment and purchase value grows 82.84 percent, and QRIS transaction value grows 186.98 percent. Payments and sales are different economic objects, so these series are not competing estimates of the same quantity and are not reconciled to issuer transaction value. Their value is to show that digital traces can expand on a different trajectory from the commerce they help record, so trace abundance is not by itself evidence of measured activity.

5.5 What the preliminary evidence establishes

Taken together, the preliminary evidence establishes feasibility at three levels. The FY2023 cases show that the transaction-revenue boundary is economically large enough to matter; the longitudinal results show that the boundary is not constant through time; and the Tokopedia reconciliation demonstrates that at least some of those movements can be traced to disclosed revenue components rather than treated as unexplained residuals. These are the core empirical claims required for the proposal oral.

6. Why the Problem Matters

The problem matters because different records support different diagnoses of the same transformation. Platform revenue growth can imply a different performance story from transaction growth; marketplace-centred evidence can mislocate where national e-commerce growth actually occurs, which is what the 98.46 percent result shows; and payment growth can be mistaken for equivalent commerce growth. At the institutional level, abundant private traces still require seller identity, a reporting rule, transmission and matching before they become administratively usable, which is the sequence PMK 37/2025 sets out. The risk this thesis addresses is therefore not that these records differ, but that one is used as a proxy for another without reconciling the boundary between them.

7. Limitations and Proposal Decisions

Table 5. Inference boundaries for the proposal examination.

Evidence supports	Does not by itself establish
$W = V - R$ and E measure the transaction-revenue boundary.	Participant profit, taxable income, unpaid tax, tax evasion or missing GDP.
Evidence tiers identify how directly each series maps to Indonesia.	That direct, scope-pending and conditional observations are interchangeable or should be pooled.
Issuer revenue components can arithmetically reconcile selected divergences.	A causal treatment effect of incentives, monetization or accounting choices.
BPS and PMK document wider activity and reporting architecture.	A firm-level causal effect or a completed compliance / tax-revenue effect.

Table 5 summarizes the inferential boundaries that govern the proposal. Measurement, accounting reconciliation and institutional interpretation are kept separate so that evidence from one layer is not used for claims belonging to another.

Two sample decisions remain for the advisor / committee: whether Blibli 3P Retail and Bukalapak Group are admissible in the final longitudinal core, and whether Grab and Shopee should remain sensitivity evidence only. The proposal exposes those decisions rather than treating them as already approved.

8. Next Step

After the proposal examination the approved sample boundary will be frozen and the requested checks completed during October-November 2026, with the full manuscript rebuilt around that boundary before the final defence. Robustness, ASEAN/payment extensions, detailed source reconciliation, and the complete verification ledger remain available in the repository and advisor data package rather than occupying proposal space. Table 6 sets out that schedule; the empirical backend is already assembled, so what remains is sample admission, requested analysis, and manuscript construction.

Table 6. Planned work to the final defence.

Period	Planned work
September 2026	Advisor review of the empirical package; proposal oral examination; freeze the sample-admission rule and the checks the committee requests.
October-November 2026	Complete requested reconciliations and source concordance; extend sensitivity to the approved tier boundary; freeze thesis tables and figures.
Before the final defence	Rebuild the manuscript under the approved sample; retain supporting payment, ASEAN and global modules only where they advance the final argument.

Appendix A - data sources by evidence class

Table A1. Source lineage by evidence class.

Tier	Series	Source of V	Source of R
Direct Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Direct, scope-pending	Blibli 3P Retail	Global Digital Niaga prospectus and results releases	Same issuer filings
Direct, scope-pending	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Conditional reconstruction	Grab Indonesia	Derived from disclosed Indonesia revenue and Group monetization rate (Form 20-F)	Grab Form 20-F
Conditional reconstruction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate (third-party)	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023, 2024; BPS directorate presentation	-
Regulatory	PMK 37/2025; DJP	Ministry of Finance of the Republic of Indonesia	-

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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